

Dodgeball Throw-Attempt Detection

Report

Gary Ferguson

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What this is. The report on the build: what was made, what it scored, where it broke, and which parts of the design the measurements overturned. The design is `docs/design.pdf`; the shipped design of each stage is in `docs/architecture/`. Every number here is reproduced by the commands in §13 from a clean checkout.

Times are clip time, m:ss from the start of the 3.5-minute cut. Decisions the build overturned are marked ▶ with the design's version beside the built one.

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1 What was built

The footage is the second half of the 2014 World Dodgeball Championship men’s final, Canada against USA, one fixed elevated end-on camera for the whole half at 1920×1080 and 25 fps.¹ The evaluation clip is 6:00 to 9:30 of that half: one complete set, from the opening rush to the next, 3.5 minutes, 5,250 frames. USA is the far team and Canada the near one in every table below. The full half, 23 minutes and eight sets, was run unlabelled for §8.

The pipeline shipped as eleven stages, each a script writing one file that the next reads.

Stage	As shipped	Against the design
Court fit	Median background plate, HSV line extraction, homography to metres, checked against markings not used in the fit	As designed
Pose	YOLO11x-pose ² at 1920 px, every frame, once per clip, single precision	As designed; half precision benchmarked, not run (§11)
Set start	Balls laid on the centre line arm a window, a whistle inside it is the start, both teams breaking for the balls confirms it	As designed
Roster	ByteTrack ³ fragments joined by jersey number (EasyOCR ⁴ on the best crops) and side; a player is whoever is in play during the set’s core	As designed, plus the in-play test: a referee’s stripes and USA’s black print read alike to the kit rule
Set end	One side down to a single player, then the court flooding with bodies	Designed as “the last elimination”; built as a state test because the elimination itself is what the outcome stage is trying to find
Candidates	Wrist speed against the torso, normalised by depth; wind-up past the shoulder line; every window a duration converted at the clip’s frame rate	As designed
Release gate	Ball as colour at the wrists before the peak, then a chain of ball-sized blobs seen leaving the hand	As designed; the pose features for follow-through were tested and dropped (§2.2)
Destination	Contact with a player’s box where the chain reaches one, otherwise the chain’s first direction in the image against the centre line	Changed (§2.3)
Rebound	SAM 2 ⁵ seeded on the chain’s last clean blob, following the ball to the player it reaches and through, on a crop that moves with it; reports whether the ball turned there	New stage (§2.1)
Outcome	A persistent step in a side’s on-court count is a hit, or a catch if the other side gains one, attributed to the last throw at that side; the rebound splits two throws inside the lag and claims blocks	Changed (§2.1)
Evaluation	One score per level, error budget, stress, ablation, plus the set-up split of §8	As designed, plus the split

Nothing was trained. The two contracts in the design held: every derived file records the clip’s SHA-256 and refuses another cut, and what is assumed about the venue (ball colour window, court dimensions, kit colours) lives in `config/venue.toml` and nowhere else.

2 What changed from the design

2.1 Outcome from the count, the ball as witness

▷ **As designed.** *The short post-release trajectory decides outcome; the on-court count checks it.*

The chain of ball blobs dies at the target’s box. At 25 fps and twenty pixels the frame where the ball meets a player is the frame the contact and everything after it are against the body, so a dodge and a hit looked the same to the chain, and the whistle sounded for fakes too (§2.2). What was observable was that someone left. Only a throw resolving moves the count, so the outcome was read from the game: a persistent step in a side’s on-court count is a hit, or a catch if the other side gains a player, attributed to the last throw at that side inside the lag. That alone scored 13 of 22 outcomes right, and its two blind spots were the ones the design had named as the biggest structural risk: two throws at one side inside the lag, where recency picks the later one and scores two errors, and blocks, which move no count and leave nothing in the state to see.

So the ball was brought back, to break ties rather than to judge. SAM 2 is seeded on the chain’s last clean blob and follows the ball to the player it reaches and through, on a crop that moves with it. Contact is the box the ball *turns* in rather than the first it enters, because a ball passing beside a crouching player enters the box without

¹ World Dodgeball Federation, *2014 World Dodgeball Championship, Men’s Final: Canada v USA* [video], YouTube, <<https://www.youtube.com/watch?v=Spu601AZHu0>>, accessed 27 August 2026.

² G. Jocher, J. Qiu and A. Chaurasia, *Ultralytics YOLO* (version 8.4), software, 2024, <<https://github.com/ultralytics/ultralytics>>, accessed 27 August 2026.

³ Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu and X. Wang, ‘ByteTrack: Multi-Object Tracking by Associating Every Detection Box’, in *Computer Vision – ECCV 2022*, Lecture Notes in Computer Science 13682 (Cham: Springer, 2022), pp. 1–21.

⁴ JaidevAI, *EasyOCR* (version 1.7), software, <<https://github.com/JaidevAI/EasyOCR>>, accessed 27 August 2026; its detector is Y. Baek, B. Lee, D. Han, S. Yun and H. Lee, ‘Character Region Awareness for Text Detection’, in *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition* (Long Beach, 2019), pp. 9357–9366.

⁵ N. Ravi, V. Gabeur, Y.-T. Hu, R. Hu, C. Ryali, T. Ma, H. Khedr, R. Rädle, C. Rolland, L. Gustafson, E. Mintun, J. Pan, K. V. Alwala, N. Carion, C.-Y. Wu, R. Girshick, P. Dollár and C. Feichtenhofer, ‘SAM 2: Segment Anything in Images and Videos’, arXiv:2408.00714, 2024.

touching them. On the clip a ball that struck a player came back off them at $77\text{--}130^\circ$; one that carried past turned under 40° ; the threshold sits at 60° . Where two throws compete for one departure, the one whose ball turned takes it; a ball that turned and put nobody out was blocked. That took the outcome to 18 of 22 (§4). The threshold was set by eye on the clip it is scored on, from six visible hits (figure 1), so scoring it blind is the next experiment (§12).

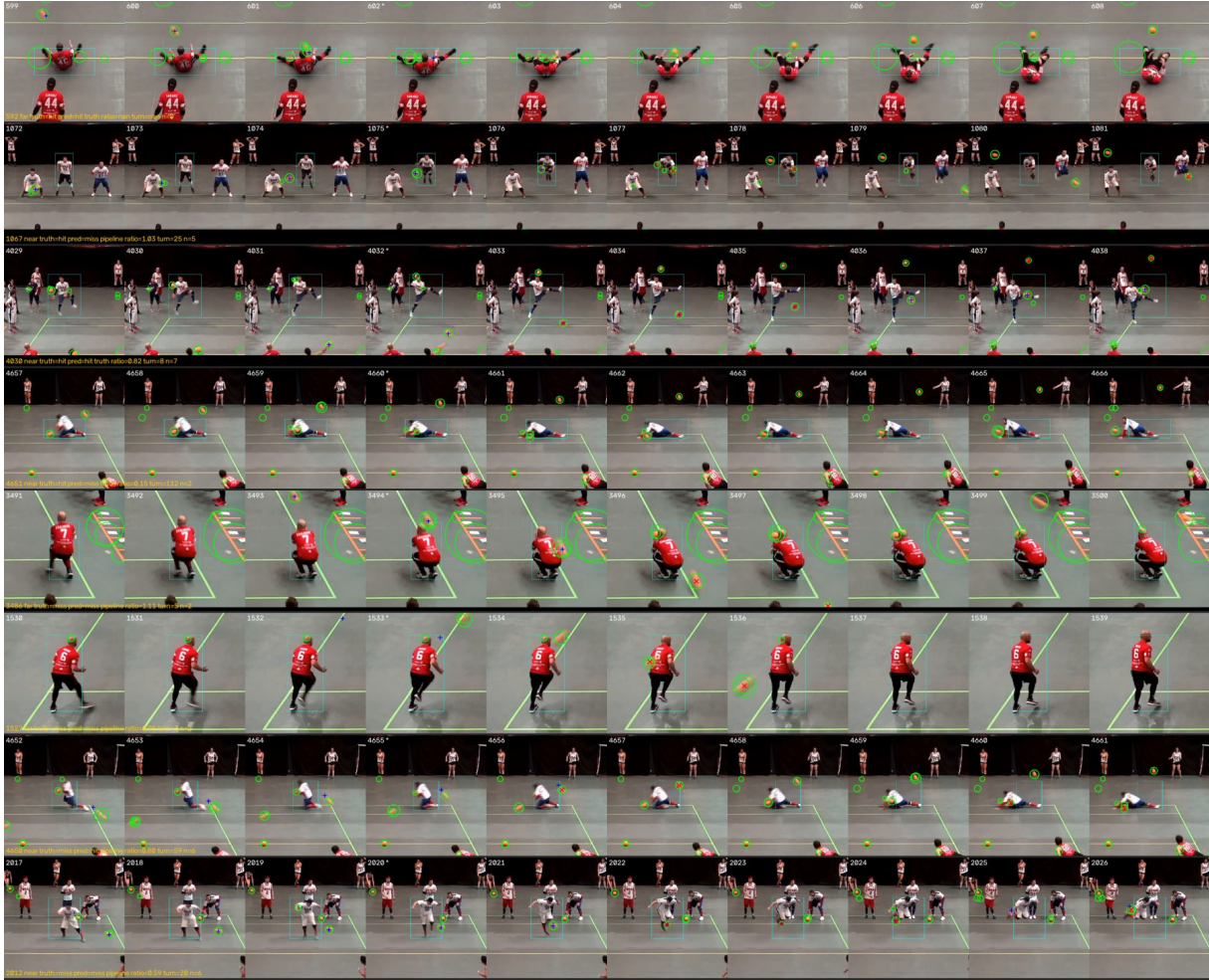


Figure 1: Four hits above four misses, ± 3 frames around the contact. The struck ball comes back off the player; the missed one carries on past. The turn angle is the one threshold the rebound stage has.

2.2 Audio at the set start only; the body cannot see a release

▷ **As designed.** Audio, the whistle band, for set start and outcome corroboration, with a vision-only against vision-plus-whistle ablation. Pose at and after the peak (follow-through, arm extension, speed decay) to be tested for feint against throw.

I thought the whistle would be richer. Referees whistle for eliminations, line violations and timeouts throughout a set, and the crowd and the shoes are in the same band. The 2.5–4.5 kHz band fired 56 times in three minutes at a floor low enough to catch quiet calls, on pump fakes as often as throws; the dozen long, loud ones were real calls and preceded far-side outs by 0.4 to 4 s, but two of them had no out behind them at all. Sixteen whistles clear the prominence floor and one of them starts the set; the rush whistle measured 37 dB over the room against 31 dB for the loudest other, which is not a margin to build on ungated. Heard only while the balls are laid out on the centre line, a whistle can mean almost nothing else, so audio is used there and nowhere else. The fusion ablation was not run because at the outcome there was nothing to fuse.

I thought follow-through might work. On the sixty labelled events and forty-five rejections, every pose feature tried for fake against release (wrist height at the peak, wind-up depth, arm extension, elbow angle, torso tilt, body speed, speed decay) sat between AUC 0.4 and 0.7; wrist height, at 0.85 on the first twenty samples, fell to 0.69 on a hundred. A fake is the same motion with the ball kept and nothing about the motion says so. The next cheapest test, the ball going dark at the hand after the peak, was AUC 0.75 and wrong in two systematic ways: a ball tucked behind a body turned to the camera goes just as dark, and a second ball in the other hand keeps the wrist lit through a real release. So the claim is made only on *seeing the ball leave*, a chain of ball-sized blobs from the hand outwards, and absence is never evidence.

2.3 Destination in the image, and by contact

▷ **As designed.** Pass or throw from the departure direction in court metres against the centre line, via the court homography.

The floor homography cannot place a ball in the air: at shoulder height it projects metres beyond the hand, and near-team throwers “stood” at court y of 11–15 m by their ball. The ball’s direction in the image is sound, though, because this camera looks along the court: the opponent is straight up the frame for the near team and straight down for the far, and a pass to a teammate goes across. Two tests replaced the metres. Where the chain reaches a player’s box, the rule’s own question is asked of that box, which needs no projection: a teammate is a pass, an opponent a throw. Eighteen of the 30 chained releases on the clip end in a box, four in a teammate’s and fourteen in an opponent’s, none the other way round. Where the chain ends in nobody, the angle between its first links and the court axis decides; the first links, because the chain’s end is whatever the ball bounced off.

2.4 Planned and not done

Per-event confidence. Every decision in the cascade is a threshold on a named quantity and that quantity is written to the timeline (ball mass at the wrist, departure distance, chain length, turn angle, how long the count step held), so a threshold sweep reads the file rather than the footage. The mapping from those margins to a $[0, 1]$ per level was not built inside the budget.

The blind second labelling pass. One annotator labelled the clip; the twenty-event double-label subset the design called for was not run. The label term of the error budget rests on the annotator’s own flags and on the game-state fold instead (§5), and the “what a second annotator would place” reading of the ± 0.25 s tolerance is untested.

3 Results

`scripts/evaluate.py` scores the timeline one level at a time against the 60 labelled events (25 fakes, 6 passes, 29 throws) at ± 6 frames (0.24 s) and $\text{IoU} \geq 0.5$ on the thrower’s box, matched same-frame same-player through the roster.

Level	Score	On
Candidate, a throwing motion here	P 84% R 93% F1 88%	67 predictions against 60 events: 56 tp, 11 fp, 4 fn
Release frame	MAE 1.1 frames, 44/56 within 2	matched events
Team	100%	56 matched
Release, fake against released	88%	56 matched: fakes right 21/23, releases right 28/33
Kind, fake / pass / throw	86%	56 matched
Throw detection, throw claimed as positive	P 73% R 76% F1 75%	29 truth throws
Outcome, hit / catch / block / miss	82%	22 matched throws; 59% by the count alone
Efficiency	near 5/17 = 29% (truth 4/15 = 27%) far 1/13 = 8% (truth 2/14 = 14%)	per team

Kind, truth ↓	fake	pass	throw	Outcome, truth ↓	hit	catch	block	miss
fake (23)	21	2	0	hit (6)	5	0	0	1
pass (6)	1	5	0	catch (1)	0	0	0	1
throw (27)	4	1	22	block (3)	0	0	2	1
				miss (12)	1	0	0	11

The timeline is `data/timeline/wdbf2014_final_h2_set2.json`: every proposal with frame, team, participant, box, released, kind, outcome and the evidence behind each claim (ball mass before the peak, departure distance, chain length and path, contact, turn angle, the count step that resolved the outcome). Dropped proposals carry why. Detection by kind, with each kind as the positive class: fake F1 78%, pass 71%, throw 75%.

4 What went wrong

Candidate misses (4). A throw whose wind-up peak landed twelve frames after the labelled release (the annotator’s note calls it late); the two fakes made with no ball, which the ball gate drops before anything is measured and the harness reports apart; and a throw at 1:49 whose label sits on a second, empty detection of the thrower, so the real thrower’s proposal, with a textbook departure, counts as a false positive against it. That one is the identity layer’s.

Candidate false positives (11). Three are the follow-through of a motion already matched, twelve to fifteen frames on: the ± 6 tolerance splits a wind-up peak from a release peak. One is the wrong-track thrower above. The rest are a ball wound up and not thrown at a moment the annotator did not call a fake: a crouch to dodge with the ball raised, a blocking arm past the line. The loose proposer was doing its job and the annotator’s negatives are unlabelled.

Releases called fake (5). A far-court throw whose ball moved under a diameter a frame; a throw noted *hard to see* whose ball the mask never got at the hand; a hand-over the annotator was unsure was a pass; a throw a hair under the departure floor (0.22 of the scale against 0.25); and the far hit at the set’s end, released in a scrum and

never seen leaving. None is a threshold away. **Fakes called released (2)**, both read as passes: a ball that split into two colour components in the fingers at the whip, one of which seeded a chain; and a fake that turned into a hand-off to a teammate. Two more kind errors, a hard throw that reached a teammate’s box first (throw read as pass) and a pass that left the hand too slowly for the chain (pass read as fake), complete the 8 of 56.

Outcome (18 of 22 right; 13 by the count alone). The rebound witness took five of the count’s nine errors: at 0:43, two near throws ten frames apart with one far departure, split by the ball that turned (77°) against the one that passed straight through; the blocks at 0:58 and 1:16, turns of 118° and 129° with no count step behind them, now claimed as block; and 1:48, a hit on a player already out, which the far departure at 1:56 falls back to once the two spurious near throws after it are seen to carry on. The four left are three shapes, all in the attribution and none in the detection, since every count step on the clip is explained by some throw:

- *The set-ending double*, 3:06. Two near throws a frame apart, a miss then a hit, whose balls both go through the same far player. The rebound follows both and reads 6° and 34°, a graze under its threshold, so recency keeps the later one and scores two errors.
- *A return the roster never saw*, 1:47. Two far catches (1:47, 1:49) put two far players off and the count shows both departures (1:53, 1:56), but only one near player walking back on (1:58). The second departure has no return to pair with and falls through to a hit on the latest near throw, a held-ball false positive at 1:50, so the catch scores miss and near efficiency gains a hit that never happened. Nothing downstream can recover it.
- *A block the ball did not show*, 2:09. The third block’s track dies inside the player’s box before any turn is seen.

The count witness’s blind spots are now the graze and the roster’s.

5 Error budget

scripts/error_budget.py reads the same matching per team and separates the model’s error from the labels’.

	near (Canada)	far (USA)
Truth efficiency	4/15 = 27%	2/14 = 14%
Predicted	5/17 = 29%	1/13 = 8%
Denominator: throws claimed	12 true, 5 matched no event, 0 fakes, 0 passes	10 true, 3 matched no event
Denominator: true throws not claimed	3 (1 called fake, 1 pass, 1 no proposal)	4 (3 called fake, 1 no proposal)
Numerator: hits	3 right, 1 missed, 2 invented	1 right, 1 missed, 0 invented
Model: paired bootstrap, 95% band on predicted minus truth	−23 to +27 points	−23 to +3 points
Labels: every uncertain event decided the other way	25% to 33%	13% to 14%
Sampling: Wilson 95% on the truth itself	11% to 52%	4% to 40%

Two readings. The near number is right for the wrong reasons: five spurious throws in the denominator and two invented hits in the numerator cancel to within two points. And the clip’s own sampling interval is wider than any pipeline error: with fifteen throws a side, this set’s efficiency says little about the team’s. The metric needs a match rather than a set, and the pipeline exists so that a match can be scored (§8).

Label uncertainty was small by the annotator’s own flags (three events) but two unflagged notes mattered more: the throw at 0:58 “could be a super narrow miss or a hit” and the one at 0:59 “could be block or catch”. Both were decided by folding the labels against the on-court count, which also found a hit at 1:48 on a player already out (recorded with eliminated: false, WDBF rule 19.1)⁶ and turned 0:59 from catch to block because no count moved. The fold is a second annotator that cannot see the ball.

6 Stress

scripts/stress.py 480p crf40 drop2: every pixel-reading stage rerun on a degraded copy of the clip; the court homography, set boundaries and labels carried over as deterministic transforms of the source (scaled, or remapped to the kept frames). Tolerance stays ±0.25 s.

Condition	Clip	Cand. F1	Release	Kind	Outcome	Throw F1	Pose-only recall	Efficiency near / far
source	1080p, CRF 16, 25 fps	88%	88%	86%	82%	75%	98%	29% / 8%
480p	854×480, same CRF	66%	90%	82%	47%	67%	88%	25% / 20%
crf40	1080p, x264 CRF 40	51%	76%	76%	67%	55%	72%	38% / 14%
drop2	every second frame, 12.5 fps	60%	93%	85%	50%	64%	57%	22% / 8%

⁶ World Dodgeball Federation, *WDBF Dodgeball Rules 2024* (2024), <<https://worlddodgeballfederation.com/wdbf-content/uploads/2024/03/WDBF-Dodgeball-Rules-2024.pdf>>, accessed 27 August 2026. Quoted by rule number in docs/reference/wdbf-rules.md.

Truth efficiency is 27% / 14% in every row. Release, kind and outcome are on matched events, whose number falls with candidate recall. The conditions were chosen to break three different things, and they did.

Frame drop looked like a units bug and was not. The release chain barely noticed half the frames (release 93%, kind 85% on matched events); the loss was at the wind-up detector, whose windows were frame counts and whose wrist-speed score was displacement per frame. Restating every window as a duration and the speed as per-second, which is how the pipeline is now written and changes nothing at 25 fps, did not recover it: pose-only recall went from 78% to 57%. A throw's whip lasts under 100 ms, so at 12.5 fps the peak is under-sampled and no rescaling of a per-frame score reproduces the 25 fps threshold. The detector's sensitivity is a property of the frame rate it was tuned at; a second rate needs its own score floor, or a detector that models the motion rather than thresholds a peak.

480p broke the ball and identity. Pose still proposed 88% of the events, but the ball at the wrists was dropped twice as often (34 “no ball in hand” against 20; an 8 px ball is a few pixels of hue), the roster fragmented, and the on-court count picked up nine steps no throw explains (none at source), which took the outcome from 82% to 47%, three misses read as catches off phantom returns. The rebound held up better than the count: the ball was still followed at 8 px and two of the three blocks were still seen.

Heavy compression broke pose itself. Blocking artefacts fired the wrist-speed detector, 137 proposals at 31% precision, and pose found 72% of the events. The gates then worked on what they were given (76% release, 76% kind, outcome 67% on the dozen throws that survived) but the denominator collapsed, near 8 throws against 15 and far 7 against 14, and the efficiency that came out is a number on half the throws. CRF 40 is past the cliff; where the cliff is between 16 and 40 was not measured.

The identity layer is rerun in every cell, so a cell's outcome number is partly the roster's; a mode that carries the source roster across would separate the cascade's limits from identity's.

7 Ablation

scripts/ablate.py: one pipeline run scored three times with the later stages withheld. The candidate matching is identical in every row; what moves is what each stage claims about the same motion.

Stage	Claims	Predictions	Throw P	Throw R	Throw F1	Fake F1	Pass F1	Kind
pose only	wind-up from pose; every proposal a throw	100	28%	97%	43%	0%	0%	47%
+ release gate	ball in hand, ball seen leaving; released means throw	67	61%	79%	69%	78%	0%	79%
+ destination	pass or throw	67	73%	76%	75%	78%	71%	86%

The release gate was worth 33 points of throw precision and cost 18 of recall, the five releases it calls fake. The destination test bought 12 more points of precision for 3 of recall, and is the only stage that finds a pass. The pose-only row is the simple baseline the brief asks for: a single-signal detector that calls every motion a throw is 43% F1 on throws and cannot see a fake or a pass.

8 The metric at scale

The metric's natural follow-up is whether *how* a throw is set up changes whether it converts. Two splits the timeline supports (scripts/tactics.py): **coordinated**, an *attack* of two or more same-team throws released within 300 ms, the tactic against a lone dodger who can only dodge one ball; and **fake-led**, same-team fakes in the 6 s before a throw. Coordination is scored per attack rather than per throw: two balls for one out is the tactic working, and per throw it would read as one for two. On the labelled set the bins hold one and two events, so the whole second half (23 minutes, eight sets, no labels) was run through the pipeline to see what holds over 240 predicted throws.

	Team	All throws	Solo attacks	Coordinated	Fakes before: 0	1	2+
labelled set, truth	far	2/14 = 14%	2/12 = 17%	0/1	2/9 = 22%	0/3	0/2
labelled set, truth	near	4/15 = 27%	3/13 = 23%	1/1	2/5 = 40%	2/8 = 25%	0/2
same set, predicted from the half	far	2/11 = 18%	2/11 = 18%	—	0/3	1/2	1/6
same set, predicted from the half	near	5/18 = 28%	4/13 = 31%	1/2	3/11 = 27%	2/4	0/3
whole half, predicted	far	47/114 = 41%	43/104 = 41%	4/5 = 80%	25/66 = 38%	17/32 = 53%	5/16 = 31%
whole half, predicted	near	23/126 = 18%	21/101 = 21%	2/11 = 18%	14/65 = 22%	6/42 = 14%	3/19 = 16%

Three readings, all with the outcome level's 82% attached.

- **The pipeline reproduced itself across cuts.** The labelled set scored from the whole-half run (set 3 of 8) landed where the clip run did: near 28% against 27% truth, far 18% against 14%. Set boundaries, roster and outcomes were all re-derived from 23 minutes of footage with no labels involved.
- **USA converted twice as often as Canada across the half,** 41% against 18%, and seven of the eight sets have the far side ahead. That is the match-level statement the metric exists to make, and it is not visible in one set.
- **Coordinated attacks: 4 of 5 far, 2 of 11 near,** against solo rates of 41% and 21%. The one coordinated attack in the truth set is the set-ending double on the last player standing, the situation the tactic exists for; five and eleven attacks cannot separate a tactic from noise (the far figure was 2 of 5 before the rebound stage moved two outcomes), and the half says only that the question is askable. Fake-led throws showed no consistent direction.

The half also showed the pipeline's limits at scale: 36 count steps no throw explains (eliminations whose throw the cascade missed or called fake), and identity fragmenting over 1,402 tracks, with the far team's numbers read on some sets and not others, which the outcome fold survives because it counts bodies, not names.

9 What the numbers cannot say

- **Recall is measured against a truth set the candidate stage seeded.** Every event was labelled from a proposal (accept, reject or adjust), so candidate recall would be 1.0 by construction unless proposals were missed; the stage was built loose for that reason (100 proposals for 60 events), and the number is better read as the precision of a recall-tuned stage. One annotator; the blind second pass was not run (§2.4).
- **The ball is colour.** An HSV mask with a hue floor one step above the red jersey, ball-sized components only. It cannot see a ball against the red team at the far baseline, and it cannot tell one ball from another when two cross; the rules that untangle a crossing are geometric.
- **The outcome is the count's, with the ball as witness.** A count step is attributed to the last throw at that side unless a ball was seen to turn at a player; a turn nobody left for is a block. The rebound's one threshold was set by eye on the clip it is scored on, from six visible hits, and a graze under it reads as a pass-through. catch is the count's alone.
- **Identity is one clip deep.** Seven players named of twelve; the far team is largely unnamed, its numbers unreadable at 90 px. Attribution is by track and side, which is what the metric needs; player-level attribution is reported where the number is known and no further.
- **One camera, one venue.** The court fit, the ball colour and the kit rules are all this footage's. Nothing here has seen a second venue.

10 Ground truth

Amount	One clip, 3.5 min, one complete set; 60 closed events: 25 fakes (2 with no ball), 6 passes, 29 throws (7 hits, 2 catches, 5 blocks, 15 misses); 105 proposal reviews, 18 with notes
Rule	docs/labeling-guide.md, written to be handed to a second person. Anchor on the release frame; thrower clicked at release, target at the end; team from the court half; kind by destination; ambiguous cases flagged <code>uncertain</code> with a note; <code>release_visible</code> , <code>outcome_visible</code> and <code>ref_signal</code> recorded so label uncertainty is separable from model error
Where it is uncertain	Three flagged events (a very close miss, a hand-over that may be a pass, a fake that became a dodge); two notes on unflagged events (0:58 hit or miss, 0:59 block or catch); two labels corrected by the game-state fold (0:59 catch to block, 1:48 hit with <code>eliminated: false</code>), each with the reason in its note
Storage	Boxes are stored by value in source pixels, never as track ids, so the truth survives a change of detector. Every label file records the clip hash
Tool	tools/labeler/: proposals shown as rings, one keypress to accept, reject or classify, player keys mapped through the roster, every box snapped to the same pose run the pipeline reads

The design estimated 50 to 70 throws and 10 to 20 feints in a set; the set held 29 throws, 6 passes and 25 fakes. Fakes were twice as common as planned for and passes rarer, which is why the pass column carries the widest interval of any level.

11 Compute, measured

Batch. One laptop RTX 4080 at an 80 W cap; the whole clip in about twelve minutes, of which pose is ten.

Stage	Measured, 3.5 min clip	Note
Pose, YOLO11x-pose at 1920 px, single precision	9.0–9.4 fps, 10 min	The design budgeted 5 min at half precision (table below); the pipeline ran single precision
Court fit	seconds	once per clip
Identity: tracking and OCR	24 s	over precomputed detections
Set start, set end	seconds	
Candidates	3 s	arithmetic over the pose run

Release gate	52 s	one sequential decode of the clip around every proposal
Rebound	~3 s per throw that reached a player	SAM 2 large on a moving crop, a model load per segment
Evaluation, error budget	1 s	
Training	none	

Half precision was benchmarked on 250 frames of the clip after the results were frozen:

Setting	fps	Keypoints against single precision
FP32, one frame at a time (as run)	9.0–9.4	
FP16, one frame at a time	18.0	median 0.43 px, p95 0.56 px; 4 of 9,256 detections differ
FP16, batch of 8	7.3	same as FP16
FP32, batch of 8	6.8, out-of-memory warning at 5.7 GiB	median 0.02 px

Half precision doubles throughput for a sub-pixel change in keypoints, and the design’s budget assumes it. Batching is slower: one 1920 px frame already saturates this GPU, and batching adds memory and CPU preprocessing. The first run of the day reads about 5 fps because the GPU idles at 210 MHz until warm.

12 Next experiment

Score the rebound witness blind. It was built, and its one threshold set, on the clip it is scored on, from six visible hits (figure 1); the second set of the half is where it holds or does not. Three things to read off that run. Whether the turn threshold holds: the set’s last hit is a 34° graze that reads as a pass-through, and a second set says whether grazes are the exception. The tracks that die inside a box (the block at 2:09, which has no answer): the tracker loses a ball that drops straight down behind a body, and a second segment seeded on the colour mask at the feet is the patch to try. And the fast far-side throws where the tracker leaves the ball before it arrives: an optical-flow bridge across the lost frames from the last good position, which cost nothing on this clip and will on a faster one.

Further out. A second venue should cost a config change and a retune: the colour window, court dimensions and kit vocabulary are in `config/venue.toml`, the court fit checks itself against markings it did not fit to, and every input is keyed to the clip’s hash, so the pipeline fails loudly at a new venue rather than adapting to it, which is the right first failure. A learned ball detector is the next model and was not built: the event needs the ball only at the thrower’s wrists and along its first metre of flight, which colour gives at this camera, and six identical balls in constant hand-off would make global tracking a linking problem needing fine-tuning and association work that the release gate does not need. It becomes worth it when the colour mask is the limit: the far baseline at 480p, or a venue whose kit shares the ball’s hue.

13 Reproducing

Python 3.12, ffmpeg, ffprobe and yt-dlp on PATH, a CUDA GPU for pose (CPU works, slowly). Footage and pose runs are never committed; labels, court fit, set boundaries, roster, candidates and the timeline are.

```
python -m venv .venv && .venv/bin/pip install -r requirements.txt
.venv/bin/python scripts/download_weights.py          # yolo11x-pose (~113 MB), sam2_1 (~224 MB)
scripts/download_footage.sh && scripts/make_clip.sh    # WDBF 2014 final, 2nd half -> 6:00-9:30 clip
.venv/bin/python scripts/run.py data/footage/wdbf2014_final_h2_set2.mp4 --offset 360
make evaluate    # every level          make budget    # error budget
make stress     # the three conditions  make ablate    # the cascade withheld stage by stage
make tactics    # the set-up split      make test      # every suite
```

14 Assistance and sources

Claude Code was used throughout as a pair: drafting modules and tests against a plan agreed first, running the evaluations, and writing the architecture docs as each stage shipped. Every design decision was taken in conversation and is recorded with its rationale in docs/; every label was placed by hand, and both documents were written by me with the assistant as editor.

Bibliography

Baek, Y., Lee, B., Han, D., Yun, S. and Lee, H., ‘Character Region Awareness for Text Detection’, in *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition* (Long Beach, 2019), pp. 9357–9366.

JaidedAI, *EasyOCR* (version 1.7), software, <<https://github.com/JaidedAI/EasyOCR>>, accessed 27 August 2026.

Jocher, G., Qiu, J. and Chaurasia, A., *Ultralytics YOLO* (version 8.4), software, 2024,

<<https://github.com/ultralytics/ultralytics>>, accessed 27 August 2026.

Ravi, N., Gabeur, V., Hu, Y.-T., Hu, R., Ryali, C., Ma, T., Khedr, H., Rädle, R., Rolland, C., Gustafson, L., Mintun, E., Pan, J., Alwala, K. V., Carion, N., Wu, C.-Y., Girshick, R., Dollár, P. and Feichtenhofer, C., ‘SAM 2: Segment Anything in Images and Videos’, arXiv:2408.00714, 2024, <<https://arxiv.org/abs/2408.00714>>, accessed 27 August 2026.

World Dodgeball Federation, *2014 World Dodgeball Championship, Men’s Final: Canada v USA* [video], YouTube, <<https://www.youtube.com/watch?v=Spu601AZHUo>>, accessed 27 August 2026.

World Dodgeball Federation, *WDBF Dodgeball Rules 2024* (2024), <<https://worlddodgeballfederation.com/wdbf-content/uploads/2024/03/WDBF-Dodgeball-Rules-2024.pdf>>, accessed 27 August 2026.

Zhang, Y., Sun, P., Jiang, Y., Yu, D., Weng, F., Yuan, Z., Luo, P., Liu, W. and Wang, X., ‘ByteTrack: Multi-Object Tracking by Associating Every Detection Box’, in *Computer Vision – ECCV 2022*, Lecture Notes in Computer Science 13682 (Cham: Springer, 2022), pp. 1–21.